16

The nature of the Indian Economy

¼Part 1 - Agriculture Sector½

In order to understand the Indian economy, we had classified it into three sectors — Agriculture and allied, Industry and Services. We can understand the changes that have taken place in the Indian economy in the last 60 years on the basis of this classification. In the initial stages of development of an economy, agriculture and allied sector have been the most important sector. The kings and feudal lords were able to establish their own armies due to the taxes imposed on the produce from the agriculture sector. At the same time cities grew, dependent on agricultural production, which were populated with large number of traders and craftsmen. As markets spread and goods were exchanged, more people started working in the industry and services sector. In spite of these changes, agriculture still remained the most important sector in society from the perspective of production and livelihood.



Figure 16.1 Fertile crop

After the Industrial Revolution, a new system of manufacturing spread throughout the developed countries. Factories were now being setup for the manufacturing of goods. A large number of people who earlier worked in the agricultural farms now started working in factories. People were now consuming factory produced low cost goods. Gradually industry sector became most important sector for large scale production and source of livelihood. We will read more about this in the history chapters.

In the developed countries, the highlight about these changes was that production as well as sources of livelihood changed simultaneously. Lesser number of people worked in the agricultural sector and the number of people working in the industries increased. From the historical perspective, this change has always happened simultaneously in production and livelihood. Is there a similar pattern of change in India or is it different? We will discuss this further in this chapter.

Agriculture and Allied Sectors

India is a developing country where more than half the population is still dependent on agriculture. At the time of independence, around 72% of the people were employed in the agricultural sector. At that time, agricultural sector contributed to around 55% of the Gross Domestic Product (GDP). 60 years

later, even today, agriculture is the primary source of livelihood for many people. Agriculture and allied sector such as dairy production, fishery and forestry provides livelihood to 53% of the people. Even then, today this sector contributes to merely 15% to the Gross Domestic Product.

Complete the table given below on the basis of figures given in the above paragraph:

Year	Contribution of agriculture to GDP	Contribution of agriculture to livelihood
Year 1950-51 (at the time of independence)		
Year 2009-2010 (60 years later)		

In view of the table given above, what are the changes that have taken place in the agricultural sector?

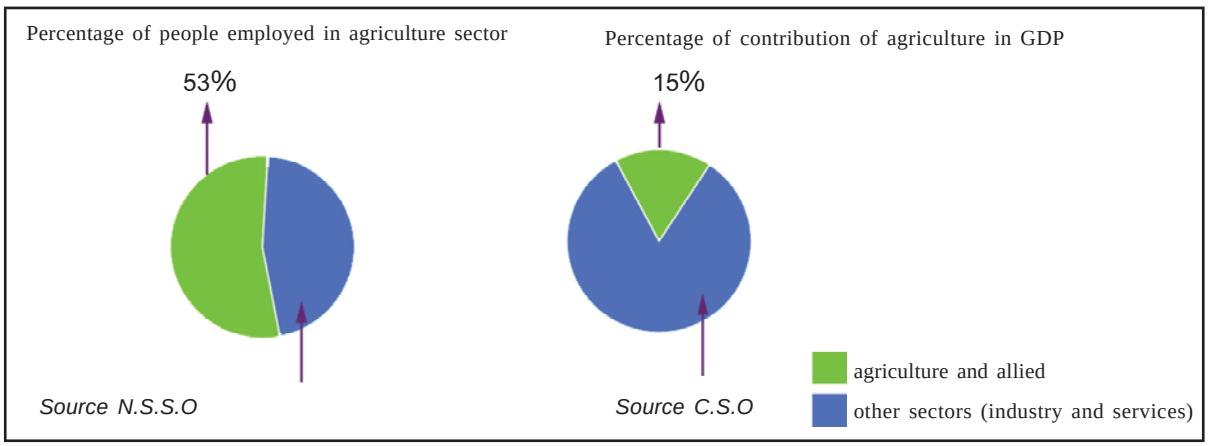
Project work:

Find out the total number of people who are employed (partially as well as fully employed) in your area.

Even though the percentage contribution of agricultural sector in the Gross Domestic Product of India has decreased in the last sixty years, large number of people still don't find employment in industry or service sector. Therefore, these people are still dependent upon agriculture for their livelihood. If some of these people work in any other sector rather than agriculture, it will not impact the overall agricultural production, that is, there will not be any decrease in the total agricultural production. If there are more people than are required in any sector of an economy, it leads to disguised unemployment. Let us understand disguised unemployment through the following example.

Imagine that there are 6 adult members in a farmer family and they have only four acres of land. These 6 people work in the farm and produce 40 quintals of paddy. Two of these members get employed in another sector and leave the agriculture. Now, four people work on the same farm to produce 40 quintals of paddy. Therefore, the agricultural productivity of these two members was in fact zero. Their going into another sector did not impact the total agricultural production. The presence of more than required number of people for this agricultural work shows disguised unemployment. In

Year 2009-10



Pie chart 16.2: population and its contribution to GDP

fact, their away going to work elsewhere benefitted their family because of addition income from these two members. As a result, this would increase the total income of the farmer family.

We will now look at the pie charts given below to understand the contribution of Indian population to the Gross Domestic Product.

Complete the pie chart and draw conclusions from it.

Have you seen any examples of disguised unemployment around you?

Situations like disguised unemployment where people have insufficient work can also be seen in cities. Discuss.

Challenges Faced by Agriculture Sector

Dependency on Monsoon and Water Conservation

Even today, most part of Indian agriculture is dependent on monsoon. Normal and timely rainfall benefits the agriculture sector. But, excessive rain and droughts disrupt the agricultural processes. Crop diseases, insects and pest, hailstorms etc. not only affect the agricultural production but also make it difficult for the farmers to recover the total cost incurred in farming. Farmers are constantly struggling with uncertainties due to unexpected changes in weather as well as the pressure to repay the loans taken from lenders.



Pie chart 16.3: destruction of crops due to hail storm

Farmers have to plan many things in order to deal with unexpected weather changes. These problems can be solved by precautionary measures like crop insurance, accumulation of grains, choosing crops on the basis of weather forecast. However, proper implementation of these measures is still a big challenge for the Indian farmers.

Today, more than 45% of the agricultural land is irrigated using different methods of irrigation. Even then, a large part of agricultural farming depends on the monsoon. It is these dry areas, where irrigation is not easily available, that water conservation is needed the most. The farmers in these areas usually grow crops like barley, gram, toor, jowar, soyabean, cotton, ground nut etc. but due to excessive use of underground water, the water level has gone down in these areas.

For instance, a person dug a tube well for his domestic and irrigation requirements 20 years ago. At that time, he had found water at the depth of 150 feet. Five years later, another villager dug a tube well, close to the previous bore well, and got water at 250 feet. The first bore well gradually dried up as the level of underground water being recharged every season slowly went lower. Three years later, the first person along with and other villagers stating digging bore wells to a the depth of 300 feet. If this

competitive digging and extraction of water goes on for a long time, it will be difficult to find any underground water at all.



Picture 16.4 : a dried up farm due to no rain

Therefore, we need to seriously consider this situation of depleting ground water levels and look for alternative solutions. For water conservation, rain water should be collected in stop dams, deep wells, Medhbandis (an earthen structure on a field's edge to prevent water from flowing out). This may prevent the decrease in water levels. People have started adopting these methods suggested by land and water scientists in some villages and cities.

What methods for grain banks could small farmers adopt to provide a sense of security during a drought.

Will crop insurance be successful in India where a large number of farmers are small and marginal? Discuss.

Project work –

You may have seen various techniques of water conservation being implemented in your area. Check whether they are being implemented & working properly or not. Write a report.

Restoring the Fertility of Land

We studied in class 8 that the green revolution had started in 1960s. Use of high yielding variety of seeds, chemicals fertilizers, pesticides and irrigation facilities helped the farmers to increase the yield. Farmers started using larger quantity of chemical fertilizers and pesticides for their crops in order to get higher yield. Food security and storage of grains was now possible and drought situations could also be controlled.

There are many microorganisms in the soil that help in maintaining the level of nutrients in the soil and keep it fertile. However the chemicals in fertilizers and pesticides, lying unused in the soil, kill these microorganisms. Due to the absence of these microorganisms, the soil loses its nutrients and therefore its fertility.

Even though the farmers are aware of that fact that the soil loses its fertility due to excessive use of chemical fertilizers and pesticides, they continue to use these pesticides and fertilizers to protect their crops from pests and insects. They feel that there's no alternative to protect their earnings. This also increases the total cost of production of the crop. We need to bring some fundamental changes in the methods of farming such as use of organic fertilizers and mixed and multi-crop farming. This is effective only if done by all or most farmers in a region. This will retain the fertility of the soil and reduce the total cost of production.

It is usually seen that all the farmers in an area plant the same crop in a particular season. Farmers have observed that by doing so there's a risk. All the crops in an area get destroyed when there are any unforeseen changes in weather. But the crops can be protected from such destruction through mixed and multi-crop farming. For instance, farmers could plant wheat, gram, masoor and mustard etc. in different farms during the rabbi season. Even if there are some unexpected weather changes, only some of the crops would suffer and the rest would still be safe. In dry regions, crops that need less water such as barley, millet, green gram and urad etc. could be produced. The biggest challenge for agriculture sector is finding ways to encourage multi-crop farming with the help of different government schemes and plans.

Organic Farming: The Experience of a Farmer

Natwar bhai is a farmer who lives in Narishu village in Cuttack, Odisha. He is a retired teacher and has been practicing organic farming for the last 20 years. He feels that organic farming can produce the same amount of crop as that from high yielding varieties. In fact, some varieties of crop that he grows yield over 20 quintals per acre. Natwar bhai does not use chemical fertilizers and pesticides in his farms. Instead, he uses natural and organic fertilizers and pesticides such as cow dung. Even though this technique requires much more effort, he has to spend less money since purchased inputs are few.



Picture 16.4 : Organic Farming

Natwar bhai was earlier a 'modern' farmer, lured into it by officials and traders, using high-yielding varieties, chemical fertilizers, and pesticides. One day, while watching a labourer spray Carbofuran (a highly toxic pesticide), he was horrified to see him stagger and collapse. Rushed for treatment, the worker survived, but not Natwar bhai's faith in the new agriculture. especially after the labourer told him: "I could not breathe, my head was reeling". After, having buried the remaining stock of Carbofuran in a pit in his fields, Natwar bhai "saw dead snails, snakes, and frogs floating in the water that had accumulated there; I immediately wondered what would be happening to the earthworms and micro-organisms that I knew kept the soil alive." Natwar bhai switched to organic farming completely.

- Adapted from 'A new rice every day?, Ashish Kothari, The Hindu, December 9, 2012'

What is organic farming? Discuss with your teacher.

Can we produce the same quantity of crop using organic farming as we get by using chemical fertilizers?

How will organic methods of farming be effective for small scale farming? Discuss.

Do farmers use the toll free number 1800-180-1551 for advice on farming? Give examples to explain.

Uneven Distribution of Land

We have seen that it is important to restore the fertility of the soil. This natural resource would help to maintain the yield of crop in the long run. On the other hand, farm land that is available for farming is

mostly distributed unevenly. Let us look at the example of a village to understand a realistic situation. This village has 450 families of farmers. Out of these, 60 families have more than 2 hectares of land. These 60 farmers are medium and large farmers. There are 240 families of small farmers who have less than 2 hectares of land each. This small piece of land is passed on among the members of the family from one generation to the other. Due to this, farm size becomes smaller and all members of these families are not able to earn their livelihood from agriculture. Many members of these families often look for employment elsewhere in other sectors, but that is not easily available. Then there are 150 families of farmers who do not own any land. They are landless. These families find work only for some part of the year. They don't find any social support or employment opportunities in the non-agriculture sector. This group is the most deprived group in the village.

Let us look at the table given here to understand more about distribution of land between small, medium and large farmers in India.

S No	Category of Farmer	Area of Land farmed by family for each category	Percentage of farmers among total farmers	Area of land cultivated in total farmland
1.	Small farmers	Less than 2 hectares	85%	45%
2.	Medium and large farmers	More than 2 hectares	15%	55%

Source – Agricultural census 2010-11

Note – The cultivated land can be farmer's own land or leased from another person.

After looking at the table above, could we say that the land has been distributed unevenly? Discuss this in class.

Marketing of Agricultural Produce

Farmers generally keep a part of the produce for themselves and sell the remaining part in the market. But usually, they are unable to get the best value for their produce due to the presence of layers of middlemen in the market. These middlemen buy the food grains at a low cost from the farmers and sell them to others at much higher rates.

In order to resolve this problem, the government now fixes the Minimum Support Price (MSP) for different agricultural products and is also making efforts to strengthen the farmers markets or Mandis. In these Mandis the farmers can sell their agricultural produce, through open auction and don't have to pay middlemen.



Picture 16.4 : Mandi

The agricultural produce is auctioned in these government Mandis after assessing the quality of the

commodity. The auctioned rates of the agricultural commodities cannot at less than the Minimum Support Price. If the government had not organized these Mandis, the farmers would have been forced to sell their produce in the open market at much lower prices and middlemen would have made all the profit. The Minimum Support Price means the minimum purchasing price of different crops declared by the government. No one can buy the produce at a price lower than the Minimum Support Price from the farmers. Therefore, sometimes the government itself makes arrangements, through village cooperative societies, for the purchase of crops so that farmers can get this declared price.

Where do farmers sell their produce in your area? Do they get fair prices for their produce? Discuss in the class and write a report.

Project work:

1. **Observe the farmers market/Mandi in your area and write a report.**
2. **Prepare a report about land distribution in your village or a nearby village with help from village officials or others.**

Role of Credit in Farming

In order to do farming, a farmer needs to buy seeds, manure and hire tractors for tilling, arrange for irrigation etc. Farmers do not have sufficient money to buy all these. So, they have to borrow money or take a loan to prepare for farming. This loan or borrowing is known as credit. This has to be returned after the harvest.

In India, farmers get credit in two ways: 1. Institutional credit 2. Non institutional credit.

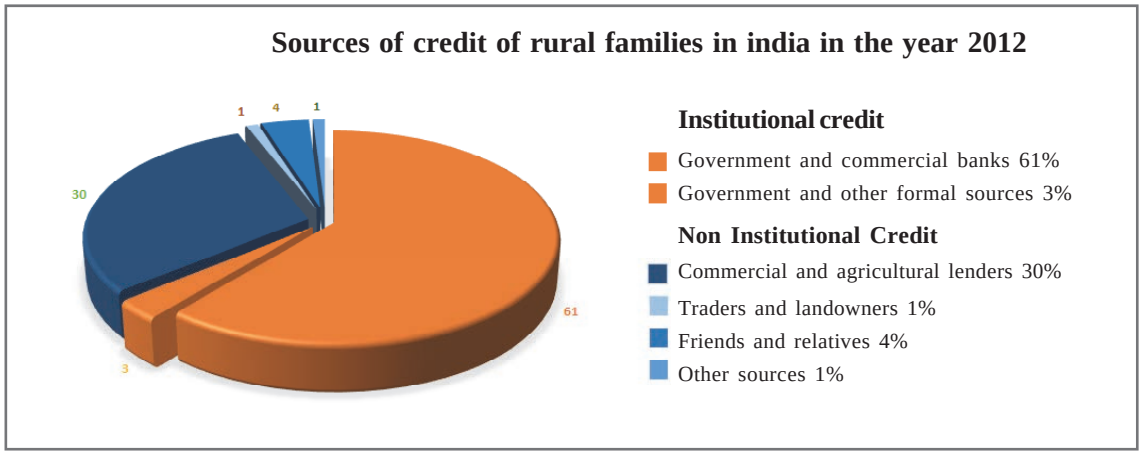
1.Institutional Credit – This is a credit facility provided to the farmers by government institutions, cooperative societies or banks. With the help of this credit facility, farmers can avail facilities like loans at low interest rates. For example use of Kisan Credit Cards, or loans for farming equipment at subsidized rates & loans for purchasing seeds, fertilizers etc. This credit facility is known as institutional credit because they have to follow the rules and regulations laid out by government.

In India, usually medium and large farmers have access to this credit facility because small or poor farmers are usually unable to furnish the collateral security required for credit. Loans are not provided without some security so that the institutions can recover their money in case of default. Moreover, there are legal documents and formalities required to get credit. Therefore, small or poor farmers don't have access to facilities like institutional credit.

2. Non Institutional Credit – Farmers who are unable to procure institutional credit have to borrow money or take loans from traders, large farmers, money lenders, friends and relatives. The lenders take advantage of this and give loans to such farmers at very high interest rates. These credit dealings are carried out on the basis of mutual understanding and oral promises. There is no paper work, legal formality or government rules involved in this. Therefore, it is known as Non Institutional Credit.

After studying the distribution of agricultural credit in India, we understand that large farmers can easily procure institutional credit whereas poor farmers who don't have money and other facilities for farming don't easily get institutional credit.

We can understand the sources of credit for the rural families in India in the year 2012 by looking at the following pie charts. Through these pie charts, we can also find out the percentage of rural families who procure institutional and non-institutional credit.



Pie chart 16.7: source NSSO report 2014

Differentiate between institutional and non-institutional credit.

What are the reasons why large farmers get institutional and small farmers get non institutional credit.

Is there a way in which poor farmers can easily get loans from banks without any security collateral? Discuss in the class.

Non Agricultural Job Opportunities

At the time of independence, farmers produced food grains mainly for their own consumption and sold a limited portion of the produce in the market. However now farming has been commercialized. Farmers buy equipment needed for farming such as tractors, tube wells, harvester and also purchase inputs such as electricity, fertilizers, seeds, etc. As we mentioned earlier, they need credit for this. A large portion of the total produce is now sold in the market, partly to repay back the credit and collect money for their family requirements.

Other allied sectors related to agriculture such as dairy production, fishery and forestry etc. fulfill the other needs of people. Most people in rural India are either small farmers or labourers. They usually don't find work in the agricultural sector throughout the year. As a result, these small and poor farmers work on the fields only during the important periods of farming such as tilling of land, sowing of seeds, and harvesting of crops. In their free time, they do other odd jobs such as labour work, selling vegetables in the local markets, brick making, and construction of houses to earn their income. Even though these non agricultural jobs are increasing in number, they only provide irregular and uncertain work opportunities and do not provide stable income to the people.

Development can happen for everyone only when we attempt to find solutions to these challenges.

Exercises

1) Fill in the blanks:

a. In the ……. countries, agriculture and allied sector were most important in the initial stages.

b. At present, around ……. percent of the population in India depends upon agriculture and allied sector for their livelihood.

c. Farmers face uncertainties all the time due to changes in …….

d. In the irrigated areas, most of the irrigation is done through ……..... water.

2. Choose the correct option:

a. New techniques of manufacturing spread in the developed countries after:

i. The green revolution ii. Milk revolution

iii. Industrial revolution iv. None of the above

b. At present, which of the following does India mostly depend on for livelihood:

i. Agriculture sector ii. Industry sector

iii. Services sector iv. None of the above

c. Which of the following is required to keep the soil fertile for longer:

i. Chemical fertilizers ii. Organic fertilizers

iii. Pesticides iv. None of the above

d. What was the contribution of agricultural sector to the gross domestic product in the year 2009-2010?

i. 17 %\ ii. 30%

iii. 45% iv. 60%

3. The livelihood of people is still dependent on agriculture sector in India. Explain the reasons for this in your words.
4. What are the reasons for the uncertainties faced by farmers?
5. Why is it important to increase irrigated areas?
6. Decreasing level of underground water is a serious problem. Write about ways to face this problem.
7. As the total production in from farms has increased there has been a simultaneous decline in fertility of the soil. What are the reasons for this?
8. Mixed and multiple crop farming are highly beneficial for farmers. Explain this statement.
9. Suggest ways in which farmers can get a fair value for their produce.
10. Based on your own experiences, discuss the steps that should be taken for the welfare of the farmers.
11. Interview one person who has taken a non-institutional loan on the basis of the following points:
    a. Purpose of the loan
    b. Relationship with the lender, if any
    c. Loan amount and interest rate
    d. Repayment of the loan taken Was there no option but to take non institutional loans? Share your thoughts about this.

Project work

Draw pictures to illustrate & explain the various methods of water conservation.

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